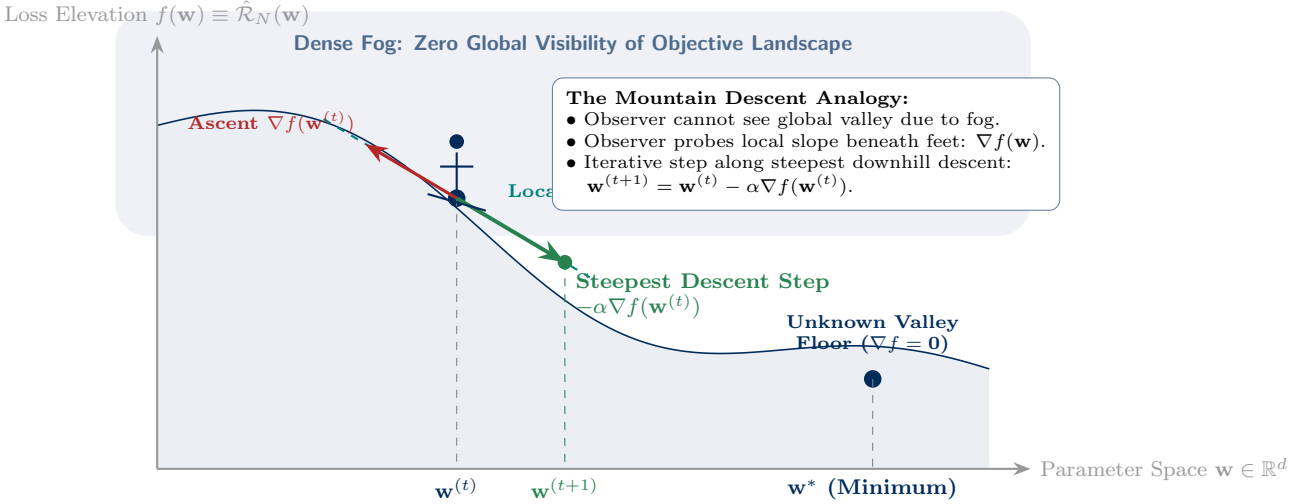

Loss Elevation $f(\mathbf{w}) \equiv \hat{\mathcal{R}}_N(\mathbf{w})$
Dense Fog: Zero Global Visibility of Objective Landscape
The Mountain Descent Analogy:
• Observer cannot see global valley due to fog.
• Observer probes local slope beneath feet: $\nabla f(\mathbf{w})$.
• Iterative step along steepest downhill descent: $\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} - \alpha \nabla f(\mathbf{w}^{(t)})$.
Ascent $\nabla f(\mathbf{w}^{(t)})$
Steepest Descent Step
$-\alpha \nabla f(\mathbf{w}^{(t)})$
Unknown Valley Floor ($\nabla f = 0$)
Parameter Space $\mathbf{w} \in \mathbb{R}^d$
$\mathbf{w}^{(t)}$
$\mathbf{w}^{(t+1)}$
$\mathbf{w}^*$ (Minimum)